

Name: _____

Date: _____

How Do Vectors Add?

SYNOPSIS

We have learned that displacement and velocity are vector quantities that require both magnitude and direction. Note that displacement is the vector "counterpart" of distance (with direction), and velocity is the vector "counterpart" of speed (with direction). However, most quantities are vectors or scalars without corresponding "counterparts."

We know how to handle scalar quantities mathematically. They are just numbers, so add, subtract, multiply, and divide like ordinary numbers. A vector, on the other hand, is like an arrow, having length and direction. Yes, they add, subtract, and multiply (but not divide, as vector division doesn't exist), but not in a simple way like numbers.

In this note, we learn how to add and subtract vectors, but we will consider multiplication later.

The vector addition rule is simple: vectors add by joining the head of one vector to the tail of another. This is known as head-to-tail addition. Subtraction is just an addition with one caveat: the negative sign flips the head and tail of the vector; that is, the vector points in the opposite direction compared to the same vector with the positive sign. In vector notation, a negative sign means the direction flips to the opposite.

If vectors are drawn connected at the tail or even separated in space, you **MUST** (re)join them head to tail (while keeping the orientation of the vectors).

Once the vectors are joined head to tail, the sum of the vectors, called the **resultant vector**, is the vector starting from the tail of the first vector and the head ending at the head of the last vector.

PERFORMANCE EXPECTATIONS

Add and subtract vectors and determine the resultant vector diagrammatically (i.e., by drawing them).

Instructions and Questions

1. What is the difference between scalar and vector quantities? Give examples.

2. How do scalar quantities add? Give examples.

3. Vectors are like "arrows." They have a length (magnitude) and point toward specific directions. And they also add, but not like numbers! But how? See below.

4. Draw the resultant vectors for a), b), and c) on the grid paper below.

Q1) When the vectors join, they make an angle. For example, a) has an angle of 180° . Using a protractor, find the angle the vectors make. What are the angles in:

a) 180° (given) b) _____ $^\circ$ c) _____ $^\circ$

Q2) What would be the length of the resultant vector when this angle is zero? _____

Q3) If these two vectors were displacement vectors, what situation does Q2 describe?

5. Draw two vectors that make an angle of zero and label them as d).

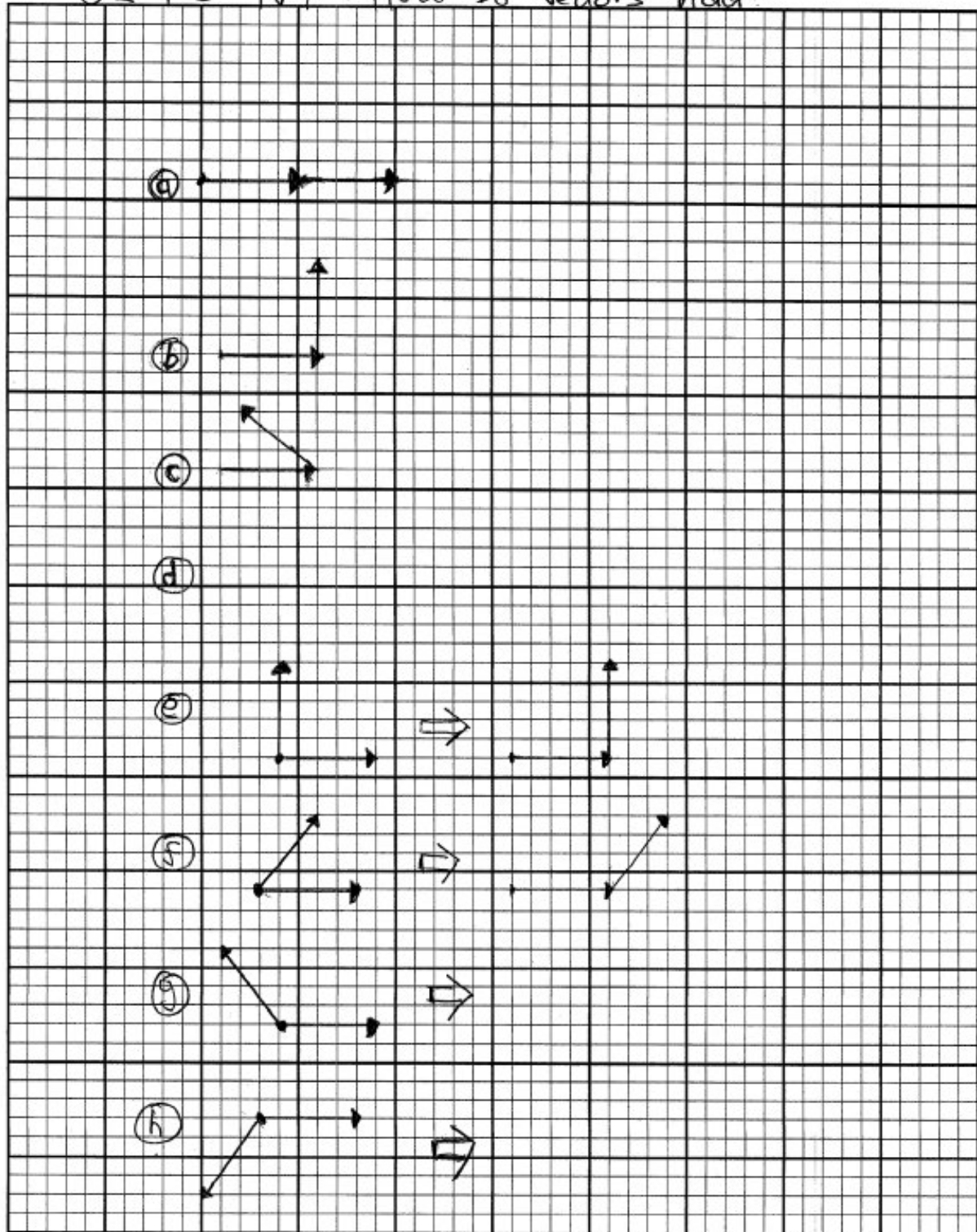
6. e) shows two vectors joined at the tail. These two might be two force vectors pulling at an object in two different directions. They can't be added when joined at the tail (or joined at the head). You must separate them and rejoin them head to tail. Complete the resultant vector for e) and f) on the right side.

7. Rejoin the vectors head to tail in g) and draw the resultant vector on the right.

8. Identify two pairs that have identical resultant vectors.

Q4) What two pairs are identical?

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Use a ruler and protractor. Draw each resultant on the grid, then answer Q1-Q4 above.